BUSI-CAD: Deep Learning for Breast Ultrasound Segmentation and Classification

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Abstract :

Breast ultrasound analysis involves both delineating lesions and interpreting image-level findings. BUSI-CAD investigates these complementary tasks through a U-Net segmentation model and an ImageNet-initialized ResNet-50 classifier on the Breast Ultrasound Images (BUSI) dataset. The segmentation workflow combines image enhancement with a binary cross-entropy–Dice objective. The classification workflow uses selective fine-tuning, training-set augmentation, and class-weighted optimization to distinguish benign, malignant, and normal images. The recorded experiment achieved 84.10% peak classification validation accuracy on 195 images, with a macro F1 score of 0.8296 at the same epoch. The final segmentation model achieved a validation soft Dice score of 0.6817. Stochastic segmentation sampling and Grad-CAM provide additional tools for examining individual predictions. Developed with clinical decision support as a long-term motivation, the project establishes a documented baseline for further work on breast ultrasound analysis.

1 Introduction

Breast ultrasound presents two related computational questions: where a lesion appears within an image, and how the image should be categorized. Segmentation addresses the first through spatial predictions, while classification addresses the second through image-level scores. Studying both tasks provides a broader view of model behavior than either task alone.

BUSI-CAD implements a U-Net for lesion segmentation and a ResNet-50 classifier for benign, malignant, and normal images. The models are trained independently: segmentation masks are not used as inputs to the classifier. Their relationship is complementary rather than sequential.

The project emphasizes the experimental work required to develop these models: preparing paired images and masks, constructing a segmentation network, adapting a pretrained backbone, handling unequal class frequencies, and evaluating predictions using task-appropriate metrics. Visual analysis extends this evaluation by making individual outputs available for inspection.

Clinical decision support motivates the choice of tasks. A future system could present a candidate lesion region alongside an image-level assessment for review by a clinician. This report focuses on the modeling and internal evaluation needed to investigate that direction.

2 Dataset and Experimental Design

The BUSI dataset contains 780 ultrasound images spanning three categories: 437 benign, 210 malignant, and 133 normal images (Al-Dhabyani et al., 2020). Corresponding annotation masks support segmentation experiments. Figure 1 illustrates examples from each category and their masks.

A shared cache stores images and masks for reuse across the notebook. The segmentation experiment uses an 80/20 training–validation split, while classification uses a separate 75/25 split. The latter contains 585 source training images and 195 validation images. Both splits operate at image level with `random_state=42`.

Classification augmentation is applied after splitting so that validation images remain separate from transformed training samples. The validation subset also determines classification checkpoint selection.

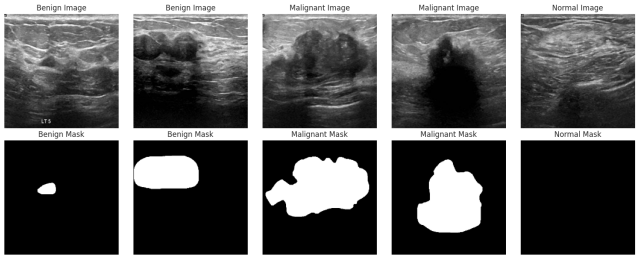


Figure 1. Examples of benign, malignant, and normal ultrasound images with their corresponding annotation masks.

The implementation uses Python, TensorFlow/Keras, OpenCV, NumPy, Pillow, scikit-learn, and Matplotlib in Google Colab. The results and figures in this report are retained from the original experimental run; the present revision updates their description and organization.

3 Image Enhancement

The segmentation workflow explores a combination of bilateral filtering, global histogram equalization, and contrast-limited adaptive histogram equalization (CLAHE). These operations address different aspects of image appearance: local intensity variation, global contrast, and contrast within smaller regions.

Bilateral filtering is applied with a diameter of 3 and intensity and spatial parameters of 75. Global histogram equalization follows, before CLAHE is applied with a clip limit of 2.0 and an 8×8 tile grid. Figure 2 shows the combined transformation on one ultrasound image.

Enhanced inputs are normalized to $[0, 1]$, replicated across three channels, and prepared at 256×256 resolution. Annotation masks are converted to binary targets.

This pipeline is used for segmentation. Classification follows a separate resized-image preparation path. The visual comparison documents the transformation itself; a controlled ablation would be needed to measure its contribution to predictive performance.

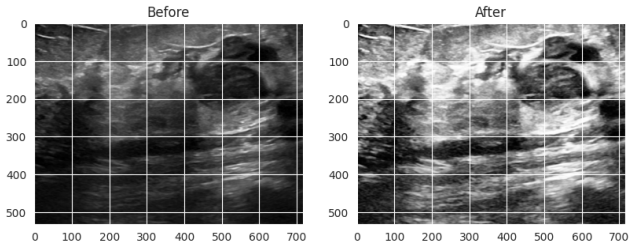


Figure 2. An ultrasound image before and after the combined enhancement pipeline used for segmentation.

4 Lesion Segmentation with U-Net

4.1 Architecture

The segmentation model follows the U-Net encoder–decoder architecture (Ronneberger et al., 2015). Four downsampling stages extract features at progressively lower spatial resolutions. Each stage contains two 3×3 convolutions with batch normalization and ReLU activation, followed by max pooling.

Feature depth increases from 16 to 128 channels across the encoder, with 256 channels at the bottleneck. The decoder uses transposed convolutions to restore spatial resolution and skip connections to combine decoder features with their corresponding encoder representations. Dropout with a rate of 0.07 is applied around the downsampling and upsampling stages.

A final 1×1 convolution with sigmoid activation produces a single-channel lesion probability map. This design combines contextual representations with access to higher-resolution features relevant to boundary reconstruction.

4.2 Training objective

Training combines binary cross-entropy (BCE) with soft Dice loss:

$$\mathcal{L} = 0.5\mathcal{L}_{\text{BCE}} + 0.5(1 - D). \tag{1}$$

The implemented soft Dice coefficient is

$$D = \frac{2\sum_i y_i p_i + 1}{\sum_i y_i + \sum_i p_i + 1}, \tag{2}$$

where y_i is the binary target and p_i is the predicted probability. The smoothing constant is 1.

The BCE component provides pixel-level supervision, while the Dice component introduces an explicit overlap objective. The model is trained for 50 epochs with Adam (Kingma et al., 2015). Loss, pixel accuracy, and soft Dice are recorded on both training and validation data.

The implementation flattens each evaluation batch before calculating Dice. Consequently, the reported value is a batch-aggregated soft Dice metric, rather than a mean of thresholded per-image Dice scores.

4.3 Segmentation performance

The final-epoch model achieved a validation soft Dice score of **0.6817**, pixel accuracy of **96.22%**, and combined loss of **0.2465**.

Figure 3 records increasing training overlap and a more variable validation trajectory. The curves provide a useful basis for examining optimization and selecting further experiments with regularization and checkpointing.

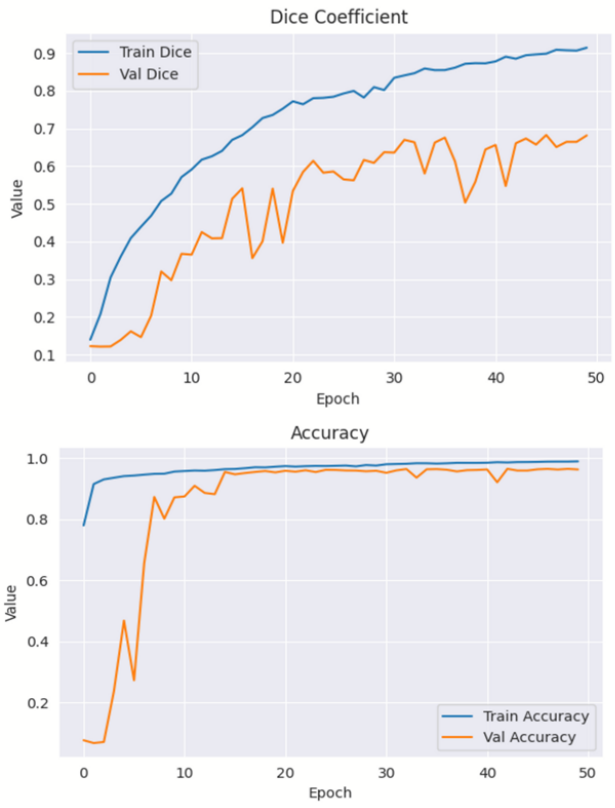


Figure 3. U-Net training histories over 50 epochs, showing loss, pixel accuracy, and soft Dice for the training and validation subsets.

Pixel accuracy includes background pixels and is interpreted alongside Dice. The result reported here belongs to the final model after epoch 50; the segmentation workflow does not restore a checkpoint selected by validation Dice.

5 Stochastic Segmentation Analysis

A single probability map does not show how predictions change under stochastic sampling. The notebook therefore includes an exploratory analysis motivated by Monte Carlo dropout (Gal et al., 2016).

For each example, 20 forward passes produce a collection of probability maps. Their mean and pixel-wise variance are calculated as

$$\bar{p} = \frac{1}{T}\sum_{t=1}^{T} p^{(t)}, \qquad T = 20, \tag{3}$$

$$v = \frac{1}{T}\sum_{t=1}^{T}\left(p^{(t)} - \bar{p}\right)^2. \tag{4}$$

The mean summarizes the sampled outputs. Comparing it with the annotation allows inspection of lesion coverage and boundary agreement.

Figure 4 retains the original qualitative examples. Prediction variance is calculated in the notebook but is not displayed in these panels.

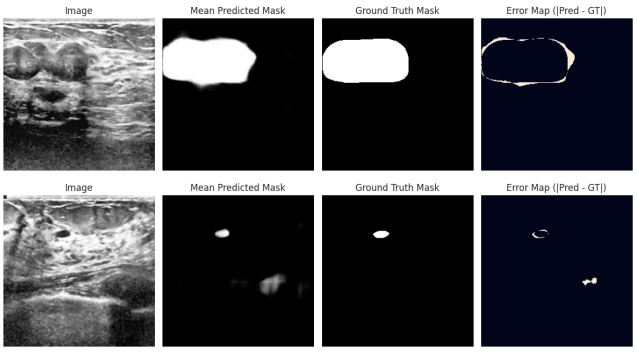


Figure 4. Stochastic segmentation examples. Each row shows the input image, mean predicted probability map, ground-truth mask, and absolute error after thresholding the mean prediction. The error panel measures disagreement with the annotation; it is not a prediction-variance map.

The current sampling call enables batch normalization alongside dropout, and the examples are drawn from the complete preprocessed dataset. The analysis is therefore exploratory. Keeping batch normalization in inference mode and evaluating variance on held-out images will allow a more specific study of dropout-induced prediction variability.

6 Classification with ResNet-50

6.1 Transfer learning

The classifier adapts ResNet-50, initialized with ImageNet weights, to the three BUSI categories. Residual connections support the use of a deep feature extractor (He et al., 2016), while selective fine-tuning adapts its later representations to the task.

The original classification layers are removed and global max pooling produces a fixed-length feature vector. The final 12 backbone layers are unfrozen.

The task-specific head contains dropout at a rate of 0.5, a 512-unit dense layer with ReLU activation, batch normalization, a second dropout layer, and a three-class softmax output. Inputs are resized to 224×224 pixels.

6.2 Augmentation and class weighting

Random horizontal and vertical flips introduce orientation variation into the training subset. Two augmented passes produce a fixed array of **1,170 training samples** from 585 source images. This array is prepared before fitting rather than regenerated at every epoch.

Class weighting adjusts the contribution of individual examples to the classification loss. Balanced weights are computed from the training labels, followed by additional multipliers of 10 for malignant examples and 4 for normal examples. These manually specified settings place additional emphasis on the two less frequent classes.

Training uses categorical cross-entropy, Adam with a learning rate of 0.0005, a batch size of 32, and 50 epochs.

6.3 Metrics and checkpoint selection

A custom callback records macro precision, macro recall, and macro F1 at each epoch. Macro averaging gives each category equal weight, complementing overall accuracy when class frequencies differ.

Table 1
Classification results at the peak-accuracy epoch. All metrics refer to epoch 28.

Metric	Value
Validation accuracy	**84.10%**
Macro precision	0.8309
Macro recall	0.8379
Macro F1	**0.8296**

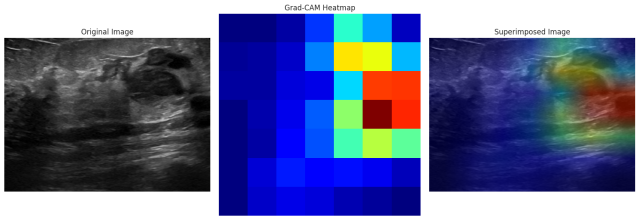


Figure 5. Preserved Grad-CAM example showing the original image, class-specific heatmap, and overlay. The source image belongs to the malignant class, while the saved prediction and corresponding heatmap are for the benign class.

The checkpoint with the highest validation accuracy is saved during training and restored afterward. The highest accuracy was recorded at **epoch 28** on **195 validation images**.

Table 1 summarizes one selected checkpoint rather than combining separately chosen metric maxima. These results establish a classification baseline for the project and provide a reference for subsequent changes to preprocessing, augmentation, and optimization.

The macro metrics summarize all three categories; they are not estimates of malignant-class sensitivity. Since validation accuracy also selects the checkpoint, the reported figures represent internal validation performance.

7 Grad-CAM Visualization

Grad-CAM relates a selected class score to spatial feature activations (Selvaraju et al., 2017). The notebook implements this analysis using ResNet-50's final convolutional block, `conv5_block3_out`.

A Keras model exposes both the convolutional activations and the classifier output. Gradients of the selected score are averaged spatially to weight the activation maps. The positive response is normalized, resized, and overlaid on the original image.

This visualization serves a different purpose from segmentation. It highlights spatial features associated with a class score rather than predicting an annotated lesion boundary. Its coarse resolution reflects the spatial dimensions of the selected feature maps.

For the retained example, `malignant (57).png`, the saved prediction is benign. The visualization also applies a ResNet preprocessing step that differs from the training input path. Figure 5 is therefore an exploratory inspection example. Aligning these transformations and regenerating the heatmaps will support a clearer comparison of correct and incorrect classifications.

8 Discussion and Clinical Motivation

BUSI-CAD brings together model construction, transfer learning, and quantitative evaluation across two related imaging tasks. The segmentation workflow combines spatial reconstruction with an overlap-aware objective. The classification workflow combines pretrained features, selective fine-tuning, augmentation, and class-weighted optimization. Custom metric tracking and checkpoint selection make the training process easier to examine and extend.

The recorded classification accuracy of 84.10% and macro F1 of 0.8296 provide a concrete baseline for three-class prediction. The segmentation soft Dice of 0.6817 supplies a separate measure of spatial overlap. Together with the retained figures, these results describe both numerical performance and individual model outputs.

The project was developed with potential clinical use in mind. A decision-support workflow could combine a candidate lesion region, class scores, and visual evidence for review by a qualified clinician. The present study contributes experimental components toward that aim; it does not establish diagnostic reliability or improved patient outcomes.

The results come from random image-level validation splits, not an independent patient-level evaluation. The next phase will focus on consistent preprocessing, refinement of mask preparation, and more detailed evaluation. Per-class classification metrics, per-image segmentation scores, and controlled comparisons of enhancement and weighting choices will help determine which changes offer measurable gains.

9 Conclusion

BUSI-CAD implements breast ultrasound segmentation and classification within a documented experimental workflow. A custom U-Net achieves a final validation soft Dice of 0.6817, while a fine-tuned ResNet-50 reaches 84.10% peak validation accuracy and a macro F1 of 0.8296 on 195 images.

The project demonstrates practical work in image preparation, architecture development, transfer learning, class-weighted training, and prediction analysis. Further development will build on this baseline through targeted experiments and broader evaluation, with clinically informed decision support remaining the longer-term goal.

Project Background and Acknowledgment

The original experiment was developed jointly by Diana Naseh and Houtan Tahanha in August 2025 as a final project for a university Data Science course. Diana Naseh prepared this revised report for the independently maintained repository. The present edition updates the description and presentation of the shared work while retaining its original experimental results.

References

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